

App Dev 2 Project Report

1. Student Details

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About Me: I am a student at IITM BS degree program. Currently in Diploma Level of the degree.

2. Project Details

Project Title: Placement Portal Application V2

Problem Statement:

To design and develop a web-based Placement Portal Application that helps institutes manage campus recruitment activities. The system allows admins to manage companies, students, and placement drives, while enabling companies to post job drives and students to apply for them.

Approach:

The Placement Portal Application was developed using **Flask** for the backend API and **Vue.js with Bootstrap** for the frontend interface. **SQLite** with **SQLAlchemy ORM** was used to manage the database, where tables were created programmatically through model definitions. The system implements **role-based access control** for Admin, Company, and Student using Flask-Security. Admin manages approvals and system data, companies create placement drives and manage applicants, while students browse drives and apply for them. Background tasks such as CSV export and reminder notifications were implemented using **Celery with Redis**. Proper backend validation and database constraints ensure data integrity and prevent duplicate applications.

3. AI/LLM Declaration

AI tools (ChatGPT – GPT-5) were used in a limited capacity for **UI-related suggestions such as Bootstrap/CSS styling and minor code suggestions** during development. The approximate extent of AI/LLM usage was **around 10–12%**.

All **backend development, database design, application logic, debugging, and integration** were implemented independently by the student. AI assistance was only used for minor guidance and formatting improvements.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
Flask-Security	Provides authentication, authorization, and role-based access control
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Vue.js	Frontend framework used to build interactive user interfaces
SQLite	Lightweight local database for storing user data
Celery	Handles asynchronous background jobs such as CSV export and scheduled tasks
Redis	Message broker used by Celery for background task processing

5. Database Schema / ER Diagram

Tables:

User — Stores authentication information and roles for all users (Admin, Student, Company).

CompanyProfile — Stores company information such as company name and HR contact, linked to a user account.

StudentProfile — Stores student academic and personal details such as branch, year, CGPA, and contact information.

PlacementDrive — Represents recruitment drives created by companies including job details, eligibility criteria, and deadlines.

Application — Stores records of students applying for placement drives along with application status.

Relationships:

One-to-One → User → StudentProfile

One-to-One → User → CompanyProfile

One-to-Many → CompanyProfile → PlacementDrive

One-to-Many → PlacementDrive → Application

One-to-Many → StudentProfile → Application

Constraint → (StudentProfile, PlacementDrive) → Application (Unique)

This constraint ensures that a student cannot apply multiple times to the same placement drive.

6. Implemented Features

Role-based authentication for **Admin, Company, and Student** using **Flask-Security**.

Company registration and approval system where companies must be approved by the admin before creating placement drives.

Student registration and profile management including academic details and contact information.

Placement drive management allowing companies to create and manage recruitment drives with eligibility criteria and deadlines.

Admin control panel to monitor students, companies, placement drives, and applications.

Application system where students can apply to approved placement drives.

Application status tracking allowing companies to update the status of applicants (Applied, Shortlisted, Selected, Rejected).

Duplicate application prevention using database **unique constraints**.

User management system allowing the admin to deactivate or blacklist users if required.

Dashboard views for **Admin, Company, and Student** displaying relevant information and system statistics.

Background job support using **Celery and Redis** for tasks such as exporting application data as CSV and sending reminder notifications.

8. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1EexgXC-nqrotzzkUPXqggLUmkAf5XKey/view?usp=sharing>